STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: FAB-04 | TAG: GRIND-PED-01

FAB-04: Pedestal Bench Top Grinders & Deburring Station

Facility Zone: Zone 5: Fabrication, Sheet Metal & Abrasive Finishing (FAB) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

CRITICAL ENTANGLEMENT WARNING: STRICTLY NO GLOVES WHILE SPINDLE / BLADE IS ROTATING

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses + Full Face Shield
2. Hearing Protection
3. NO GLOVES Near Wire or Stone Wheels (Immediate Snag Hazard)

2. PRIMARY OPERATIONAL HAZARDS

1. Workpiece wedging into tool rest gap (>1/16") causing wheel explosion
2. High-velocity wire bristles detaching from deburring wheel
3. Grinding aluminum/brass on aluminum-oxide wheel (loads wheel pores and causes violent shattering)

3. BEFORE OPERATION CHECKLIST

1. Verify tool rest gap is strictly <= 1/16" (0.0625") and upper spark tongue guard is <= 1/8" (0.125").
2. Inspect stone wheel for cracks or non-ferrous loading.

4. DURING OPERATION SAFETY RULES

1. Stand to the side of the wheel plane during startup for 30 seconds.
2. Grind on the FACE of the stone wheel only (NEVER grind on the side of the wheel).
3. Quench hot metal frequently in water pot and hold small parts with locking pliers.

5. AFTER USE & SHOWROOM CLEANUP

1. Turn switch OFF and let wheels coast down completely.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Flip center toggle switch OFF and step out of wheel plane immediately.